

Technology Stack

Date	26 June 2026
Team ID	EduGenie - Google Gemini Powered Learning Assistant
Project Name	XXXXXX
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, Jinja2 templates, CSS, JS	No build step required; templates render server-side from FastAPI, keeping the stack lightweight for a fast demo turnaround.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python 3.10, FastAPI, Uvicorn	Async-first framework with automatic OpenAPI docs, native Pydantic validation, and fast iteration for REST endpoints.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	JSON flat files (data/*.json) <i>[Specify technology]</i>	7Easily persist data that mirrors a relational schema (Users, Queries, Responses, Quizzes, Summaries, Learning Paths) without relational constraints (e.g., handles unstructured data, strong relational mapping)] <i>[Explain why this was chosen]</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Docker (python:3.10-slim) <i>[Specify technology]</i>	Containerization ensures the app runs consistently across development and evaluation environments; exposed on port 8000. <i>(e.g., easy scalability, cost-effective hosting)]</i> <i>[Explain why this was chosen]</i>
5	Version Control & CI/CD	Git, GitHub <i>[Specify technology]</i>	Standard collaborative source control for the team; no automated CI pipeline is configured yet. <i>(e.g., seamless collaboration, automated testing)]</i> <i>[Explain why this was chosen]</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Google Gemini API & local LaMini-Flan-T5 model	Combines cloud-grade reasoning quality (Gemini, with dynamic best-model resolution) with an offline-capable local model for resilience.